

Learning Reflection Journal

Workshop: Building an Image Classifier with PyTorch (Chihuahua vs. Muffin)

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Summary of Workshop Objectives and Techniques

The goal of this workshop was to build a simple image classification model using PyTorch that could distinguish between photographs of chihuahuas and muffins. Although these two categories seem very different, they often share similar colors, textures, and shapes. These similarities make them a surprisingly challenging image classification problem. This project helped me understand the basic workflow of creating and training a neural network for image classification in a Jupyter Notebook.

The workshop followed several important steps. First, the image dataset was loaded using the torchvision library. Next, the images were resized to a consistent size of 224×224 pixels. They were then converted into tensors and normalized so the neural network could process the data efficiently. After preprocessing, the training and validation datasets were loaded into DataLoader objects. These DataLoaders organized the images into batches and shuffled the training data before each training epoch.

Next, we built a fully connected neural network by creating a class that inherited from PyTorch's `nn.Module`. The model included an input layer, two hidden layers that used ReLU activation functions, and an output layer that produced predictions for the two image classes. Unlike more advanced image classifiers, this model was trained completely from scratch. It did not use pretrained models or convolutional neural networks (CNNs).

After defining the model, we selected `CrossEntropyLoss` as the loss function and Stochastic Gradient Descent (SGD) as the optimizer. The network was trained for five epochs while tracking both training and validation accuracy. At the end of training, the notebook displayed graphs showing how the loss decreased and how the model's accuracy improved over time. The final validation accuracy reached approximately 83%. These results demonstrated that even a simple neural network can learn meaningful patterns from image data.

Key Concepts Learned

This workshop strengthened my understanding of several fundamental machine learning concepts. One of the most important lessons I learned was how a neural network gradually improves through repeated training. During each epoch, the model made predictions and

compared them with the correct labels. It then calculated the error using the loss function and adjusted its weights through backpropagation. Repeating this process allowed the model to improve its performance over time.

I also gained a better understanding of the importance of preprocessing image data. Resizing every image to the same dimensions ensured consistency throughout the dataset. Converting the images into tensors allowed PyTorch to perform the mathematical operations needed during training. In addition, normalization helped stabilize the training process and improved the model's learning efficiency.

Another important concept I learned was the purpose of DataLoaders. Instead of processing every image at once, DataLoaders organized the dataset into smaller, manageable batches. This approach made the training process faster and more memory efficient. I also learned the difference between the training dataset and the validation dataset. The training dataset was used to teach the model, while the validation dataset measured how well the model performed on images it had not previously seen. This helped determine whether the model was learning useful patterns instead of simply memorizing the training data.

Challenges Encountered

The most challenging part of the workshop was understanding how image data moved through the neural network. Since the model used fully connected layers, each image had to be flattened into a single long vector before it could be processed. At first, I did not understand why the input dimensions had to match the first linear layer exactly. After reviewing the code and tracing the tensor shapes step by step, I gained a better understanding of how the data flowed through the network.

Another challenge was learning how each part of the training loop worked together. I had to understand the roles of the optimizer, the loss function, the forward pass, and backpropagation. This required me to read through the notebook carefully and test the code several times. Watching the loss decrease over multiple epochs helped me connect these concepts to the actual training process. It also helped me understand how the model gradually improved its predictions with each training epoch.

Insights Gained

Completing this workshop gave me a much clearer understanding of what happens during the model training process. Before this assignment, I understood neural networks mostly as theoretical concepts that I had learned about in class. Building a neural network from scratch helped me understand how each line of code contributed to the learning process.

I also realized that creating an effective machine learning model involves much more than simply defining the network. Preparing the data correctly is an important part of the process. Selecting appropriate training parameters is equally important. Monitoring the model's accuracy throughout training also helps evaluate its performance. Together, these steps contribute to building a successful machine learning model.

Even though the classifier was relatively simple, it demonstrated how a neural network can learn patterns from image data through repeated training. Completing this workshop helped me connect the concepts I learned in class to a practical machine learning application. As a result, I feel more confident in my understanding of how neural networks are built, trained, and evaluated.

Real-World Applications

The techniques used in this workshop can be applied to many real-world image classification problems. Similar machine learning workflows are used in healthcare to classify medical images and assist healthcare professionals in making diagnoses. In manufacturing, image classification models are used to inspect products for defects and improve quality control. In agriculture, these models help identify plant diseases from photographs of crops. In retail, they can automatically recognize products and organize inventory.

Personal Reflection

Overall, this workshop increased my confidence in using PyTorch and Jupyter Notebook. Building the model step by step helped me better understand concepts that had previously seemed difficult. These concepts included neural networks, loss functions, optimizers, and training epochs. Seeing each part of the code work together made these topics easier to understand.

I found it rewarding to watch the model gradually improve as the training progressed. Observing the validation accuracy increase over time showed me that the model was learning meaningful patterns from the image data. Seeing these improvements made the training process feel more meaningful and helped me appreciate the importance of each stage of model development.

This experience also taught me that building machine learning models requires patience and careful attention to detail. Even small mistakes during data preprocessing or model configuration can affect the model's performance. For that reason, it is important to understand each step of the workflow before moving on to more advanced techniques. After completing this workshop, I feel more comfortable working with PyTorch and Jupyter Notebook. I also look forward to learning more advanced image classification techniques in future assignments after mastering these fundamental concepts.

References

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press